

Rock and root: Which played more important roles in preferential flow in the rocky mountain area

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ABSTRACT

The role of rock and root structures in preferential flow within mountainous terrains remains an underexplored area in hydrological research. This study aimed to assess the influence of these structures on preferential flow in the rocky mountainous regions of North China, utilizing computed tomography and dye tracing experiments. Our key findings indicated that rock fragments content and coverage increase with slope elevation, with medium-sized rock fragments particles being predominant. Conversely, root characteristics, including quantity density, surface area density, and volume density, were found to be more abundant in the lower slope. Additionally, the staining area from dye tracing, as well as the preferential flow length index and maximum infiltration depth, exhibit a decreasing trend with increasing soil depth, while the upper slope showing significantly higher values compared to the middle and lower slopes. The partial least squares path model indicated that rock fragments were the primary drivers of preferential flow, with medium rock fragments having the most substantial impact. A threshold of approximately 6.92% coverage for coarse rock fragments was identified as the limit beyond which preferential flow is inhibited. Overall, this study provides a thorough analysis of preferential flow dynamics, highlighting the greater influence of rock structures compared to root systems in the study region, and offers critical insights for water resource management in semi-arid mountainous areas.

1. Introduction

Preferential flow (PF), a key hydrological phenomenon, refers to the non-equilibrium movement of water that bypasses a significant portion of the soil matrix, resulting in accelerated flow velocities (Allaire et al., 2009; Qin et al., 2000; Valtera et al., 2023). This process significantly affects soil water absorption, mitigates surface runoff and soil erosion (Guo et al., 2020), and promotes groundwater recharge (Guo & Lin, 2018; Ireson & Butler, 2011; Kang et al., 2023). Its role is particularly critical in mountainous regions, where water scarcity, thin soil layers, and high rock fragment content exacerbate hydrological challenges (Gao et al., 2020). Despite increasing attention, the complex mechanisms and

controlling factors of PF remain poorly understood, necessitating further research (Clothier et al., 2007).

Empirical evidence indicates that PF is governed by multiple interacting environmental factors, including soil properties (e.g., macropores), topography, and anthropogenic influences (Demand et al., 2019; Koestel & Jorda, 2014; Vereecken et al., 2019). Among these, macropores—formed by root channels and rock fragments—play a pivotal role by creating direct conduits for rapid water transport (Hlaváčková et al., 2019; Zhang et al., 2020). Specifically, roots substantially enhance PF through the formation of stable channels that retain moisture and facilitate deep infiltration (Cai et al., 2024; Zhang et al., 2012). However, roots may also impede soil permeability, particularly during initial

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growth stages or in highly porous soils (Li et al., 2023).

Beyond root systems, rock fragments play a particularly critical role in governing PF dynamics in mountainous regions. These fragments (diameter >2 mm), formed through weathering pedogenesis of fractured bedrock (Poesen & Ingelmo-Sanchez, 1992), exert multifaceted influences on PF processes (Dai et al., 2017; Demand et al., 2019). Under certain conditions, rock fragments contribute to PF pathway development (Jiang et al., 2019) by enhancing soil hydraulic conductivity and fostering macropore formation (Lv et al., 2017). However, they may also inhibit PF by reducing the effective infiltration area (Hou et al., 2023). Notably, Hou et al. (2023) demonstrated that locally effective rock fragments exhibit a dual role: they can promote PF when forming seepage paths along the soil-rock interface, while conversely depressing PF movement when their total volume fraction exceeds 30 %. This complexity is further illustrated by Wang et al. (2022), who found increasing PF frequency in deeper soil layers from upslope to downslope on forested hillslopes, with both roots and rock fragments being significant contributors. Additionally, Zhang et al. (2023) revealed that PF degree correlates strongly with macropore number rather than size. Despite these advances, our understanding of spatiotemporal PF distribution remains limited due to system heterogeneity and factor interactions. Thus, quantitative analysis of PF characteristics and their relationships with influencing factors represents a critical yet challenging research frontier for guiding ecological restoration across diverse regions.

Root systems profoundly affect soil water infiltration by forming channels and interfaces that alter soil hydrological responses, including water uptake, nutrient acquisition, solute retention, and soil conservation during root growth (Li et al., 2013; Xiao et al., 2024). Specifically, plant roots grow into soil pores to form continuous preferential root channels (Nespoulous et al., 2019; Tracy et al., 2011). Studies by Bargaes Tobella et al. (2014) and Zhang et al. (2015) demonstrate that both active and decomposed roots establish effective conduits for water and solute transport. Consistent with this, Jørgensen et al. (2002) observed more pronounced water movement and solute transport in soil profiles containing root channels compared to those without. However, Hou et al. (2024) revealed a non-linear relationship, where PF initially increases with root biomass, density, and length, but declines beyond a threshold level. Furthermore, findings from mountainous regions show inconsistent root system effects (Li et al., 2023), suggesting these hydrological impacts vary substantially across soil conditions and vegetation types. Therefore, systematic research remains crucial to elucidate the complex, context-dependent nature of root-mediated PF processes.

Current understanding of the interactions between PF, rock fragments, and root systems remains limited. Notably, Hou et al. (2024) demonstrated that PF development results from the combined influence of rock fragments and roots rather than their individual effects. In particular, their study revealed that in severe rocky desertification areas, sparse vegetation leads to underdeveloped root systems, while high rock fragment content reduces effective soil matrix space – conditions that collectively inhibit rapid water infiltration. This spatial variability is further evidenced by Mei et al. (2018), who observed macropore-dominated flow in root-rich upper slopes but rock fragment-induced lateral flow at lower slopes. To bridge the gap identified in the literature, it is crucial to explore how these factors interact within the soil matrix. This gap in research impedes a thorough understanding of hydrological processes, especially in complex mountain ecosystems where the interaction between roots and rock fragments significantly influences water flow dynamics.

Partial Least Squares Path Modeling (PLSPM) is a multivariate statistical method designed for constructing and evaluating complex causal models (Sun et al., 2023). This method effectively integrates the advantages of both structural equation modeling and regression analysis, making it particularly suitable for examining intricate path relationships (Barberán et al., 2014). Given these capabilities, we employed PLSPM to quantify the relative contributions of root systems and rock fragments in

governing PF dynamics.

Mountains, which are present across every continent, account for approximately 24 % of the land surface area of the Earth, covering nearly 36 million km² (Chakraborty, 2021). These rocky regions play vital ecological and hydrological roles, serving as biodiversity hotspots and critical regulators of global water resources. Within these contexts, PF significantly influences these regional hydrological processes. Additionally, rock fragments and plant roots are particularly influential to soil structure and hydrological dynamics in these areas. To advance understanding of these mechanisms, our study aims to: (1) characterize PF dynamics in rocky mountainous areas, (2) discuss how the rock fragments and roots influence the PF, and (3) determine whether roots or rock fragments have a more significant impact on PF. Our hypotheses are: (i) that an increase in rock fragments content can enhance PF by raising the number of macropores, but may also inhibit it when it surpasses a specific threshold; (ii) and that in mountainous areas, the beneficial effects of plant roots on PF might be diminished by the presence of rock fragments, potentially leading to adverse effects. These findings will provide critical insights into hillslope hydrology, supporting improved watershed management in mountain ecosystems.

2. Materials and methods

2.1. Study site description

The study was conducted at Henan Xiaolangdi Earth Critical Zone National Research Station, Jiyuan City, Henan Province, China, located at coordinates 35°01'N, 112°28'E. (Fig. 1). Characterized by a warm-temperate continental monsoon climate, the region has a mean annual temperature of 13.1 °C and receives an average annual precipitation of 613 mm, with a potential evaporation rate of 1611.2 mm (Zhang et al., 2021). In recent years, rainfall depths of 0–10 mm constituted over 65 % of events in this region, while 78 % exhibited intensities below 1 mm h⁻¹ and 14.2 % ranged between 1–2 mm h⁻¹. The predominant soil type in the area is brown soil with a predominantly sandy texture. The vegetation was dominated by low trees and shrubs, and the dominant species included *Robinia pseudoacacia* L., *Quercus variabilis* Bl., *Vitex negundo* L. var. *heterophylla* (Franch.) Rehd., and *Ziziphus jujuba* Mill. var. *spinosa* (Bunge) Hu ex H. F. Chow.

In May 2021, a comprehensive field survey identified two representative sloping fields within the study area. Vegetation composition was consistent across both slopes, transitioning from grasses on the upper slope, to shrubs on the middle slope, and trees on the lower slope. Based on these vegetation patterns, the slopes were divided into upper, middle, and lower sections. Soil thickness varied by slope position: 20 cm at the upper slope, 25 cm at the middle slope, and 45 cm at the lower slope. On the upper slope, the soil profile consisted of an A horizon (eluvial horizon, 0–5 cm) overlying a C horizon (parent material, 5–20 cm). On the middle slope, the profile included an A horizon (0–10 cm) above a C horizon (10–25 cm). On the lower slope, the sequence comprised an O horizon (organic horizon, 0–3 cm), an A horizon (3–25 cm), and a C horizon (25–45 cm). Slope 1 had a gradient of 18.49°, an elevation of 361.45 m, and dimensions of 85 m in length and 30 m in width. The corresponding lengths for each slope position were 40 m (upper), 20 m (middle), and 20 m (lower). Slope 2 had a gradient of 19.05°, an elevation of 360.33 m, and dimensions of 100 m in length and 30 m in width. The corresponding lengths for each slope position were 40 m (upper), 30 m (middle), and 30 m (lower) (Niu et al., 2024). Bulk density, organic matter content, texture, particle size composition, and soil depth for each slope segment on hillslopes 1 and 2 were measured and are presented in Table 1.

2.2. Sample collection

2.2.1. Field design and rock fragments sampling

For the rock fragment coverage on the slope 1 and slope 2, the grids

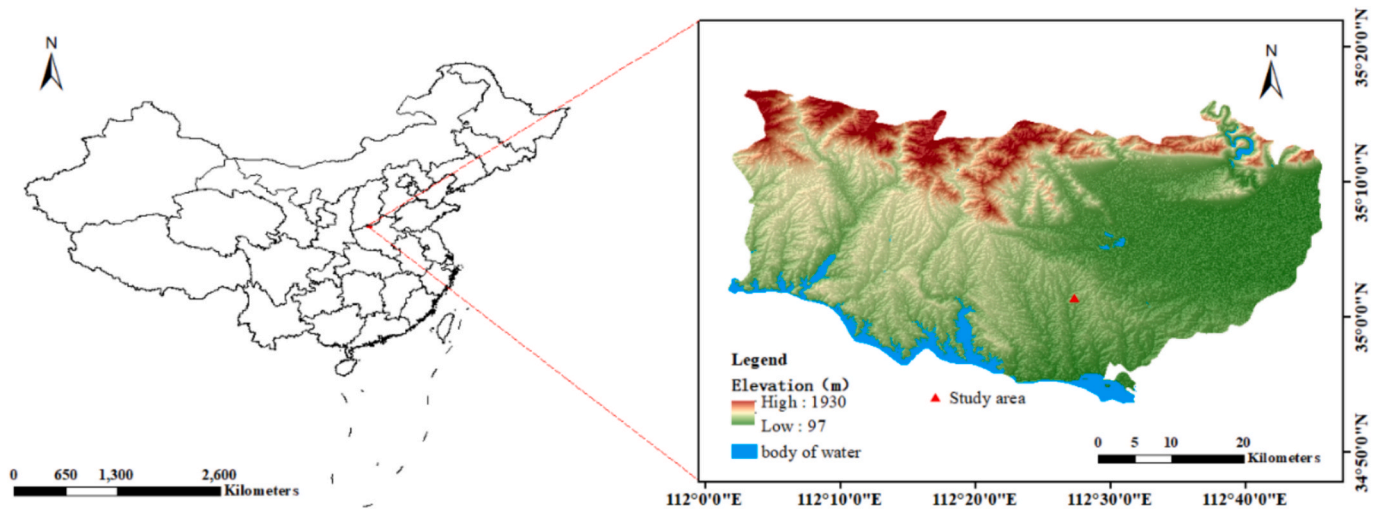


Fig. 1. Location of the study areas.

Table 1

Basic information of sample plots. TC means total carbon and TN means total nitrogen.

		Bulk density	TC	TN	Texture (%)			Soil thickness
		(g cm ⁻³)	(g kg ⁻¹)	(g kg ⁻¹)	Sand content (0.05–2 mm)	Clay content (<0.002 mm)	Silt content (0.002–0.05 mm)	cm
Slope 1	Upper slope	1.56	16.90	1.94	90.30	2.63	7.08	20
	Middle slope	1.47	28.45	2.82	89.43	2.02	8.55	25
	Lower slope	1.40	36.36	3.06	89.12	2.23	8.65	45
Slope 2	Upper slope	1.48	17.50	1.53	89.28	4.38	6.34	20
	Middle slope	1.51	16.47	1.81	90.47	2.80	6.74	30
	Lower slope	1.42	20.34	1.81	89.20	2.47	8.33	40

were partitioned based on the standard of 5-m horizontal intervals and 10-m vertical intervals, and samples were obtained at the grid points. However, some samples could not be obtained because certain grid points were distributed on exposed bedrock. Finally, 32 and 43 sample points were organized from the slope 1 and slope 2, respectively, about 10 sampling points for each slope segment. To investigate the rock fragment coverage and size on the soil surface (1 m × 1 m) of the two hillslopes, digital photography and image processing techniques were employed, following the method described by Zhu and Shao (2008). The method had a maximum relative error of $0.7 \pm 0.8\%$. At each soil sampling point on both slopes, a 1 m² area of the soil surface was selected; prior to photography, surface vegetation was removed, and the slope gradient was measured. A digital camera was then positioned at a fixed distance of 1.5 m from the soil surface to capture images of each sampling point. The images were subsequently processed using ImageJ software (Fig. S1, (Niu et al., 2023)). Specifically, a grid was overlaid onto each image, and at each grid node, the presence or absence of a rock fragment was recorded, along with its size class, following the classification system of Poesen et al. (1997). The intermediate diameter – defined as the longest diameter measured perpendicular to the greatest length – was used to determine fragment size, as most fragments larger than 75 mm were flat in shape. The lower size limit for rock fragments was set at 5 mm, since smaller fragments (< 5 mm) were assumed to exhibit physical behavior similar to sand (Poesen et al., 1997). Within each 1 m² plot, all surface rock fragments (> 5 mm) or those with less than 50 % of their volume embedded in the topsoil were collected and sorted into three size classes: 5–20 mm (medium rock fragments), 20–75 mm (coarse rock fragments), and >75 mm (cobbles). The number of fragments in each size class was recorded for every sampling site.

Furthermore, we measured the volume concentration of rock fragments at these same sampling locations for the rock fragment coverage. Soil samples were collected deep to bedrock. A soil pit with dimensions of 0.3 m in length, 0.3 m in width, and 0.4 m in depth was excavated at

each point using a small sampling box. Soil samples were collected every 10 cm depth and laid out on a canvas, and then the rock fragments were separated from fine-grained soil particles using sieves with mesh sizes of 5 mm, 25 mm, and 75 mm. The volume of rock fragments was then quantified employing the water displacement method (Chen et al., 2011). Given the porosity of rock fragments, they were encapsulated within thin plastic bags, degassed to expel any entrapped air, and subsequently submerged in water to determine their volume accurately.

2.2.2. X-ray scanning and image analyse

Undisturbed soil samples were taken near the sampling points for the rock fragment coverage and the volume concentration. Excluding excessively disturbed samples, three undisturbed soil sample were collected at each slope segment and a total of 18 undisturbed soil columns were collected. Prior to soil sampling, the surface vegetation, including dead branches and leaves, was carefully removed. Undisturbed soil samples were then collected using PVC pipes with dimensions of 400 mm in length, 160 mm in diameter, and 6 mm in wall thickness; each pipe was fitted with a round stainless-steel cutter at the bottom to facilitate penetration into the soil. The ground surface was cleaned, and then the PVC pipes were gently tapped vertically into the soil to a depth of 400 mm or until the bedrock. Once inserted, the soil surrounding the pipes was carefully excavated to allow extraction of the pipes containing the undisturbed soil samples. Finally, the extracted PVC pipes, which contained the undisturbed soil samples, were sealed with plastic wrap and padded with sponges for safe transportation to the laboratory.

The soil columns were scanned using a 128-slice helical CT scanner (InsitumCT 568, Senovatech Co., Ltd). The scanning parameters were set as follows: a slice thickness and interval of 0.625 mm, a peak voltage of 140 kV, and a current of 300 mA. The resulting image matrix resolution was 512 × 512 pixels, and the attenuation value of the obtained images ranged from −1024 to 3071 (Fig. S2). Post-scan image processing in

ImageJ software involved thresholding to differentiate rock fragments (attenuation value 1379 to 3071), root systems (attenuation value –594 to 248), and macropores (attenuation value less than –595), and then converting the images into binary format (Zhang et al., 2023). Next, the “3D objects counter” plugin was then applied to determine the number and volume of soil macropores and root systems. The “Volume Fraction” plugin was applied to determine the microporosity. Adhering to the methodology outlined by Luo et al. (2010), we calculated three-dimensional parameters, including volumetric density, number density, and surface area density, for both macropores and root systems. Furthermore, the “3D viewer” plugin was utilized to achieve a three-dimensional reconstruction and visualization of the soil macropores present within the soil cores.

2.2.3. Blue dye tracing experiments

To accurately capture PF, dye tracing experiments were conducted near undisturbed soil sample sites, with three replicate plots established at each slope segment. This experimental design resulted in nine plots per slope and a total of 18 plots across both slopes. From each plot, three vertical soil profiles were excavated, yielding a comprehensive dataset of 54 soil profiles for analysis.

Prior to the experiment, all herbaceous plants, lianas, and litter were meticulously removed from the plots. Custom metal frames, measuring 60 cm by 60 cm by 50 cm with a thickness of 0.6 mm, were vertically inserted into the soil surface to a depth of approximately 5 cm (Fig. S3). Extreme care was taken during this process to avoid any disruption to the soil structure. The final staining profiles were removed 5 cm from both the left and right sides, with an image width of 50 cm. The aim was to avoid the influence of staining edge effects on the experimental results and to ensure consistency in data analysis. The experiment was conducted on days when no rainfall had fallen the day before. We selected Brilliant blue for coloring food dye as the tracer for its low toxicity, excellent fluidity, and high visibility post-dilution and adsorption. A constant current agricultural water sprayer (3WBD, Zhongnong Fengmao Plant Protection Machinery Co., Ltd) was used for water injection, positioned 50 cm above the soil level directly above the sample box (Fig. S3). A total of 20 L of distilled water (equivalent to 5.5 mm of rainfall) was uniformly applied to a sample plot at a rate of 1.25 mm h⁻¹, simulating moderate rainfall. The preliminary experiment showed that this rainfall depth (5.5 mm) could penetrate the entire soil profile, while not caused excessive matrix flow that would stain the entire profile and make it impossible to identify preferential flow pathways. The rate of 1.25 mm/h was chosen representatively to ensure a steady and controlled infiltration process, allowing the dye to trace the flow paths effectively. The water injection was halted immediately if water accumulation was observed, resuming only after complete infiltration. To reduce the impact of evaporation, a plastic film was laid over the plots. The soil profile, stained with the dye, was excavated 24 h later.

The soil profiles were carefully documented using a digital camera, which was set up under an opaque sunshade during daylight hours. This setup minimized direct solar radiation and ensured consistent lighting (Fig. S4). First, the images were processed in Photoshop CS3 to adjust geometry and lighting. Subsequently, using MATLAB 2019, the stained and unstained areas were segmented to generate high-contrast images, and the proportion of stained soil was quantified using ImageJ. Specifically, within ImageJ, the images were converted into binary format: stained pixels were assigned a value of 255 (black), while unstained pixels were assigned a value of 0 (white) to facilitate the analysis of stained soil proportions. Additionally, the images were corrected to eliminate the influence of slope within the experimental site on the data. The number of stained pixels was calculated based on the soil surface, and vertical images were resampled at 0.01-m depth increments to determine soil depth for data presentation.

The dye coverage (D_c), length index (Li), preferential flow ratio (P_F), uniform flow depth (U_f), and maximum infiltration depth (U_{max}) were calculated to analyze the morphological characteristics of soil

preferential flow staining.

D_c is defined as the ratio of the dye-stained area to the total soil vertical profile. Following Flury et al. (1994), D_c was calculated as:

$$D_c = (D/(D + N_D)) \times 100 \quad (1)$$

where D denotes the stained area (cm²) and N_D non-stained area (cm²). Stained areas were quantified via pixel-grid matrix conversion, with the total profile width = 600 cm (600 pixels).

Layer-specific dye coverage (D_{ci}) represents the stained area ratio at depth i . D_{ci} was computed in Matlab R2008b by dividing stained pixels (0.5 mm × 0.5 mm) by total pixels per row. Data were averaged over 1-cm depth increments (20 rows) for subsequent analysis.

Li is defined as the sum of the absolute differences between consequent D_c values with depth in a profile, which describes the heterogeneity degree of the dye penetration pattern (Bargués Tobella et al., 2014). It can be calculated as follows (Hou et al., 2023):

$$Li = \sum_{i=1}^n |D_{c(i+1)} - D_{ci}| \quad (2)$$

where n is the total layer count, and $D_{c(n+1)} = 0$ was defined as the baseline for deeper strata.

Uniform infiltration depth (U_f) is the depth at which the stained area ratio is higher than 80 %, which assumed to be the infiltration front of the area dominated by uniform flow (Bargués Tobella et al., 2014). In contrast to preferential flow, U_f represents the depth of the uniform and continuous infiltration front.

P_F is the ratio of the preferential flow area to the total stained area in the soil dyeing section:

$$P_F = (1 - (U_f \cdot W)/TSA) \times 100 \quad (3)$$

where U_f is the depth of the soil profile corresponding to the matrix flow area (cm), W is the horizontal staining width of the soil profile ($W = 50$ cm); and TSA is the total stained area of the soil profile (cm²), the stained area was taken as an estimate of infiltration.

2.3. Statistical analysis

We used PLSPM to clarify the direct and indirect effects of macropores, rock fragments, and roots on the PF. PLSPM consists of two components: a measurement model and a structural model (Sun et al., 2023). The measurement model defines the relationship between latent variables and indicator variables through principal component analysis, while the structural model specifies the relationship between latent variables through ordinary least squares (Barberán et al., 2014). For each latent variable, we applied an iterative algorithm to estimate the loading values between the latent variable and indicator variables, aiming to maximize the explanatory variance of the latent variable. PLSPM was conducted using the “Smart PLS 3 (SmartPLS GmbH)”.

After Pearson correlation indicated significant linear relationships ($P < 0.05$), linear regression was performed to obtain slopes, intercepts, and 95 % confidence bands in Origin (version 2021). The peak value of the nonlinear fitting curve. Differences among slope positions were evaluated with one-way ANOVA followed by Fisher's LSD; normality was checked with Shapiro-Wilk and homogeneity of variances with Levene test, while non-normal datasets were log-transformed or analysed with Kruskal-Wallis test. All tests were two-tailed at $\alpha = 0.05$ in IBM SPSS 25 (SPSS Inc., USA). Data integration was performed using Microsoft Excel 2019. We used Origin 2021 and GraphPad Prism 9 for data visualization.

3. Results

3.1. Rock fragment distribution

The distribution of rock fragments in the study sites showed a decreasing trend across different particle sizes, from upper to lower slope positions (Table 2). Notably, the upper slope contained a significantly higher concentration of rock fragments compared to the middle and lower slopes, except for cobbles. Total rock fragment concentration varied by slope position (Table 2), with medium fragments (21.09 %) being the most abundant, followed by coarse rock fragments (10.14 %) and cobble (2.94 %). The upper slope particularly stood out with a middle rock fragments content (*MCn*) of 24.33 %, which was 1.4 times and 1.6 times greater than that of the middle (17.08 %) and lower (15.04 %) slopes, respectively. This pattern of rock fragments content corresponded with the distribution of rock grain sizes, with the upper slope also having the greatest volume concentration of rock fragments at 40.6 %, surpassing the middle slope (25.6 %) by a factor of 1.6 and the lower slope (18.5 %) by a factor of 2.2.

The surface area coverage of rock fragments decreased as the slope grade decreased, with significant variations in middle rock fragments coverage (*MCv*) and total rock fragments coverage (*TCv*) across the slope positions. As can be seen from Table 2, medium rock fragments dominated the soil surface coverage at 6.33 %, while cobbles were the least at 0.75 %. This aligns with the overall trend in rock fragment distribution, with medium rock fragments being the predominant type across the slopes. Specifically, the upper slope displayed the highest coverage (8.14 %), surpassing the middle (3.87 %) and lower (1.83 %) slopes by factors of 2.1 and 4.4, respectively. In contrast, coarse rock fragments and cobbles accounted for <1 % coverage, except for coarse rock fragments (2.61 %). Furthermore, *TCv* peaked on the upper slope (11.72 %), exceeding the middle (5.01 %) and lower (2.06 %) slopes by 2.3-fold and 5.7-fold, respectively.

3.2. Root distribution characteristics

Root densities exhibited a distinct increasing trend from upper to lower slope positions (Table 3). Both root density and root surface area density increased from the upper to the lower slope (Fig. 2), a pattern further supported by root volume density measurements, confirming denser root networks in lower slopes. In specific terms, the root density in the lower slope was approximately 56 % higher than that in the middle slope and 55 % higher than that in the upper slope. Likewise, the root surface area density in the lower slope was about 58 % greater when compared to both the middle and upper slopes. For the volume density, the lower slope showed 27 % and 33 % increases relative to the middle and upper slopes, respectively. Additionally, roots demonstrated higher

Table 2
Spatial distribution characteristics of rock fragment slope.

Particle size	Rock fragment volume content (%)			Rock fragment coverage (%)		
	Upper slope	Middle slope	Lower slope	Upper slope	Middle slope	Lower slope
Medium gravel	24.33 ± 9.28aB	17.08 ± 7.24bB	15.04 ± 5.31bB	8.14 ± 2.43aB	3.84 ± 1.56bA	1.83 ± 1.12cA
Coarse gravel	12.28 ± 10.73aC	6.43 ± 8.36bC	3.01 ± 3.55bC	2.61 ± 2.21aC	0.71 ± 0.62bB	0.23 ± 0.52bB
Cobble	3.86 ± 8.16aD	2.09 ± 9.33aC	0.46 ± 1.77aC	0.96 ± 1.84aD	0.46 ± 1.62abB	0 ± 0bB
Total gravel	40.46 ± 17.33aA	25.6 ± 16.04bA	18.5 ± 8.65bA	11.72 ± 4.27aA	5.01 ± 2.41bA	2.06 ± 1.52cA

Different lower-case letters indicate significant differences at different slope positions ($P < 0.05$), and Different up-case letters indicate significant differences at different particle sizes.

Table 3

Root characteristics of soil columns as determined using the CT images.

Slope position	Number of density (N mm ⁻³)	Surface area density (mm ⁻² mm ⁻³)	Volume density (mm ⁻³ mm ⁻³)
Upper slope	0.0019 ± 0.0005	0.028 ± 0.0112	0.0175 ± 0.007
Middle slope	0.0021 ± 0.0009	0.0292 ± 0.0166	0.0183 ± 0.0104
Lower slope	0.0024 ± 0.0008	0.0373 ± 0.0114	0.0233 ± 0.0071

surface densities across all measured soil depths.

3.3. Macropore characteristics

Macropores quantified in all the samples were reconstructed for visualization (Fig. 3). Overall, the data revealed a consistent trend of decreasing soil macropore quantity and macroporosity as slope position declines. Specifically, the soil column on the upper slopes contained an average of 923 ± 464 macropores, whereas the middle and lower slopes had averages of 795 ± 406 and 402 ± 199 macropores, respectively (Table 4). The soil on the slopes exhibited a significantly higher number of macropores, being 1.16 times and 2.30 times that of the soil on the upper and lower slopes, respectively. In terms of macroporosity, the upper slope had a rate of 0.61 ± 0.18 %, which was considerably higher than the middle slope's 0.36 ± 0.06 % and the lower slope's 0.15 ± 0.04 %, representing increases of 1.69 times and 4.06 times, respectively.

Macroporosity generally decreases with increasing soil depth, although it shows slight increases at various depths. Macroporosity on the upper slope initially increases and then decreases at soil depths ranging from 40 to 60 mm and again from 100 to 120 mm. In contrast, the middle slope exhibits a slight increase in macroporosity specifically within the depth range of 50–70 mm. The lower slope, however, does not display a significant increase in macroporosity. Overall, macroporosity on the upper slope is 1.69 and 4.07 times higher ($P < 0.05$) than those on the middle and lower slopes, respectively.

3.4. Characteristics of preferential flow

3.4.1. Area ratio of preferential flow staining in different slope positions

Dying patterns varied significantly across the three slope positions, leading to distinct differences in the staining area ratio along the soil profiles (Fig. 4). Notably, the horizontal diffusion range was most extensive in the surface layer, particularly within the top 6 cm, indicating a clear interaction between PF and matrix flow regions at this depth. As soil depth increased, PF paths become evident, with water flow patterns differing among slope positions. On the upper and middle slopes, PF paths were strongly influenced by local rock fragment distribution. The upper slope exhibited a grid-like arrangement of stained sections, whereas the middle slope displayed greater instability and pronounced finger flow characteristics. In contrast, matrix flow dominated water transport in the lower slope.

The staining area ratio exhibited a generally decreasing trend with soil depth across all slope positions (Fig. 4). On the upper slope, the staining area ratio was more than 80 % within the top 0–5 cm. Although the ratio decreased with depth, the trend was not strictly monotonic. A significant drop was observed between 6–20 cm, followed by an “S”-shaped pattern indicating a steady decline beyond 20 cm. Notably, there was a slight increase in the ratio at depths of 30, 45, and 54 cm. For the middle slope, the staining area ratio was above 80 % at 0–6 cm, then it gradually decreased, showed a slight rise at 38 cm, and finally approached zero. Likewise, the lower slope mirrored this behavior, with an initial ratio >80 % at 0–7 cm, a progressive decrease, a minor resurgence at 30 cm, and a convergence toward zero.

The average *Dc* values varied significantly among the three slope

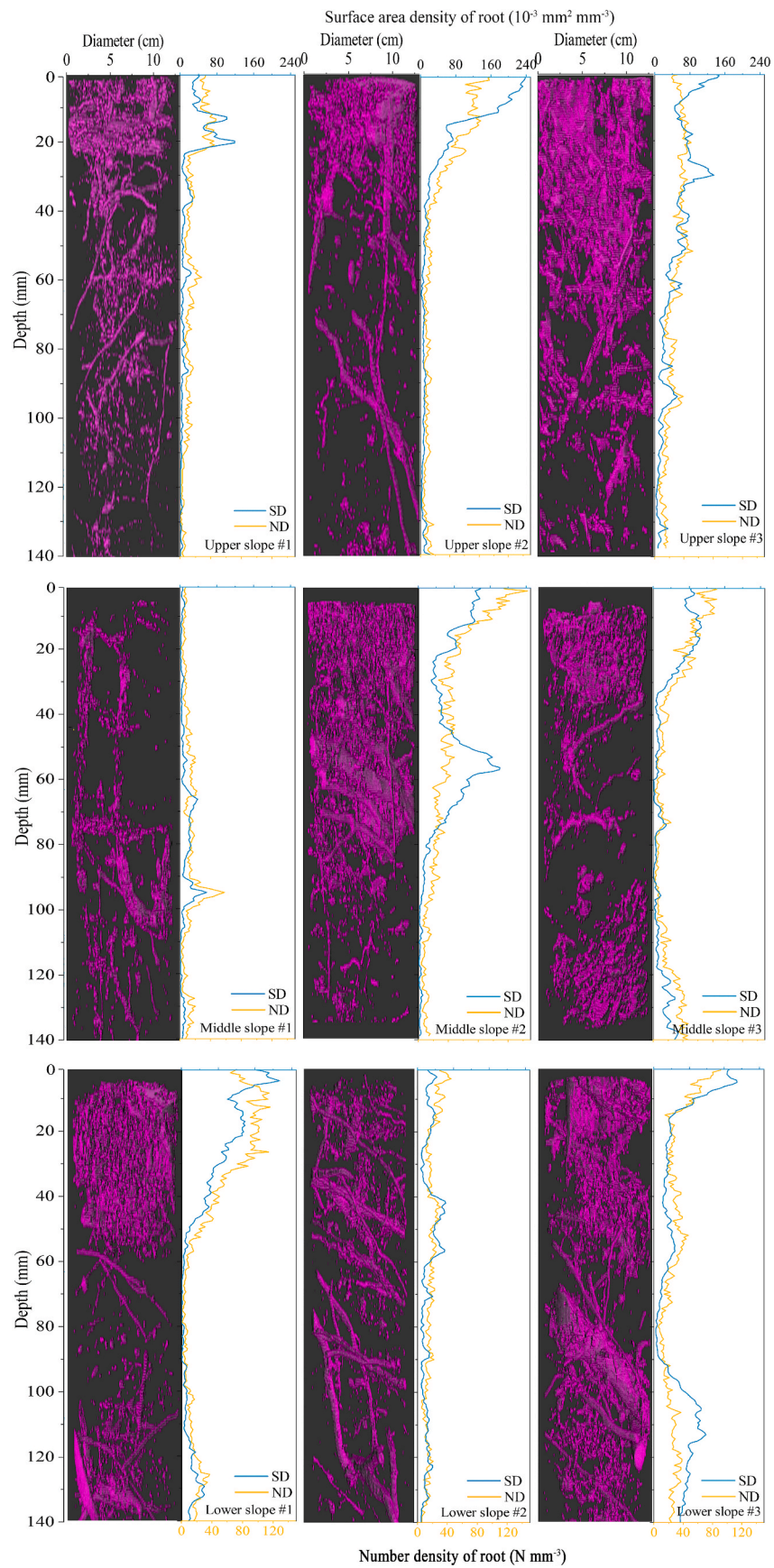


Fig. 2. Three-dimensional visualization of root networks (purple color) for the soil column, ND: number density of root; SD: surface area density of root.

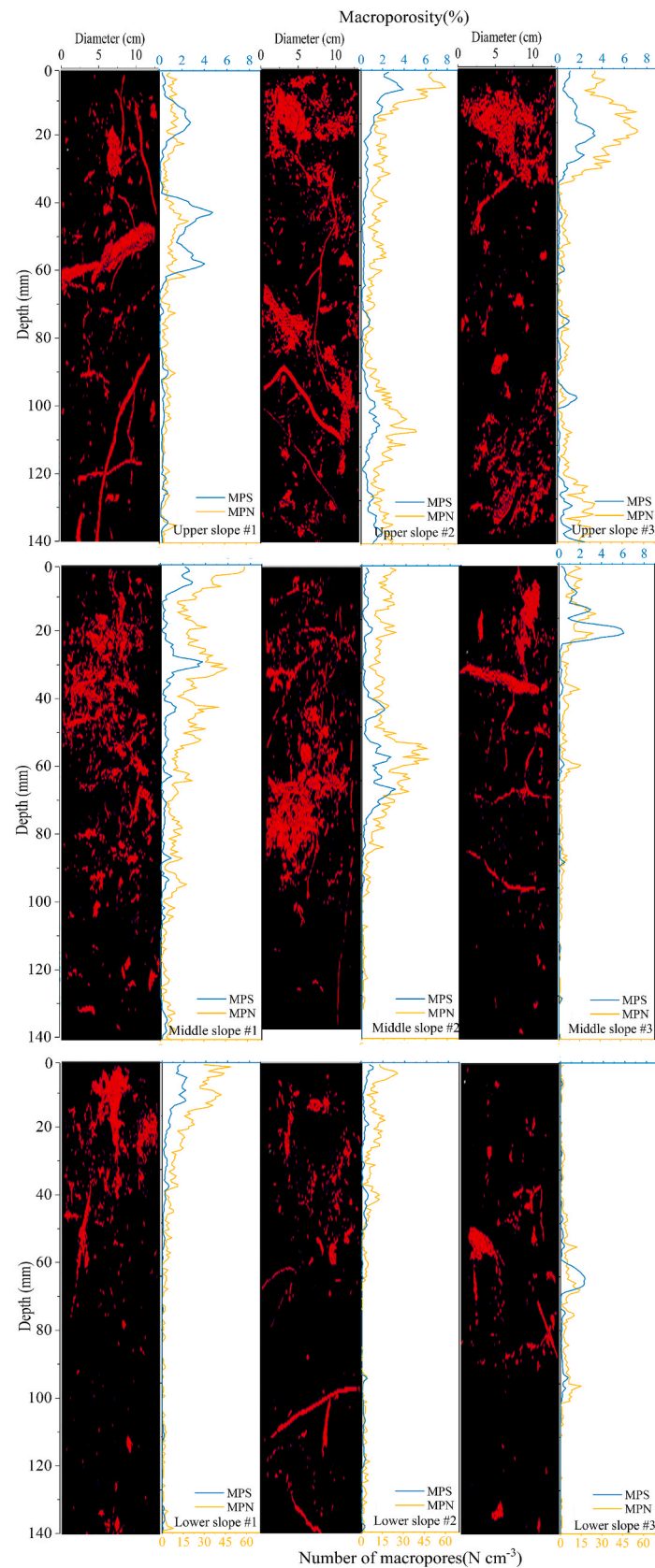


Fig. 3. 3D reconstruction of macropores (red color) in the soil samples. And characteristics of the number of macropores and macroporosity at different slope positions. MPN: number of macropores; MPS: macroporosity.

Table 4
Statistical summary of macropore parameters from computer tomography images in three slope positions.

Slope position	Number of macropores	Mean macropore diameter (mm)	Macroporosity (%)
Upper slope	923 ± 464a	2.4 ± 0.89a	0.61 ± 0.18a
Middle slope	795 ± 406a	1.79 ± 0.16a	0.36 ± 0.06b
Lower slope	402 ± 199a	1.9 ± 0.24a	0.15 ± 0.04b

Different lower-case letters indicate significant differences at different slope positions ($P < 0.05$).

positions at equivalent soil depths (Fig. 5). In the surface layer (0–6 cm), D_c values were highest, with the lower slope showing the maximum value (93.67 %), followed by the middle slope (92.5 %) and upper slope (89 %) (Fig. 5, b). Below this layer (7–20 cm depth), D_c values decreased markedly. Notably, at depths >20 cm, this trend reversed: the upper slope exhibited the highest mean D_c (6.63 %), exceeding both the middle slope (5.09 %) and lower slope (1.26 %).

3.4.2. Characteristic parameters of soil preferential flow at different slope positions

The characteristics of preferential flow parameters at different slope positions are shown in Fig. 6. The L_i , D_c , P_F , and U_{max} revealed a consistent trend of decreasing values from the upper slope to the lower

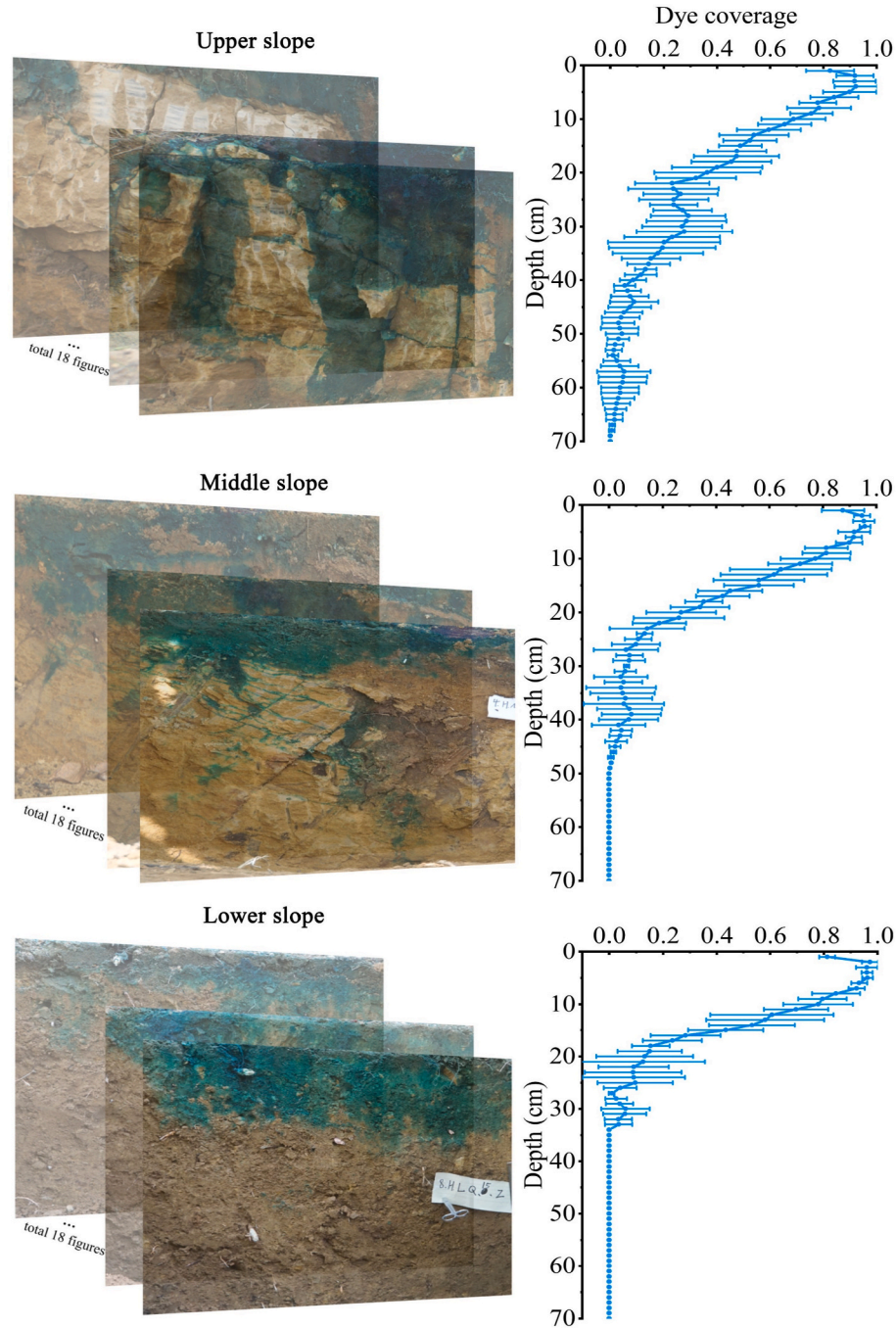


Fig. 4. Examples of stained soil profiles influenced by rock fragments and their corresponding profile of indices used to quantify the degree of preferential flow in different slope positions.

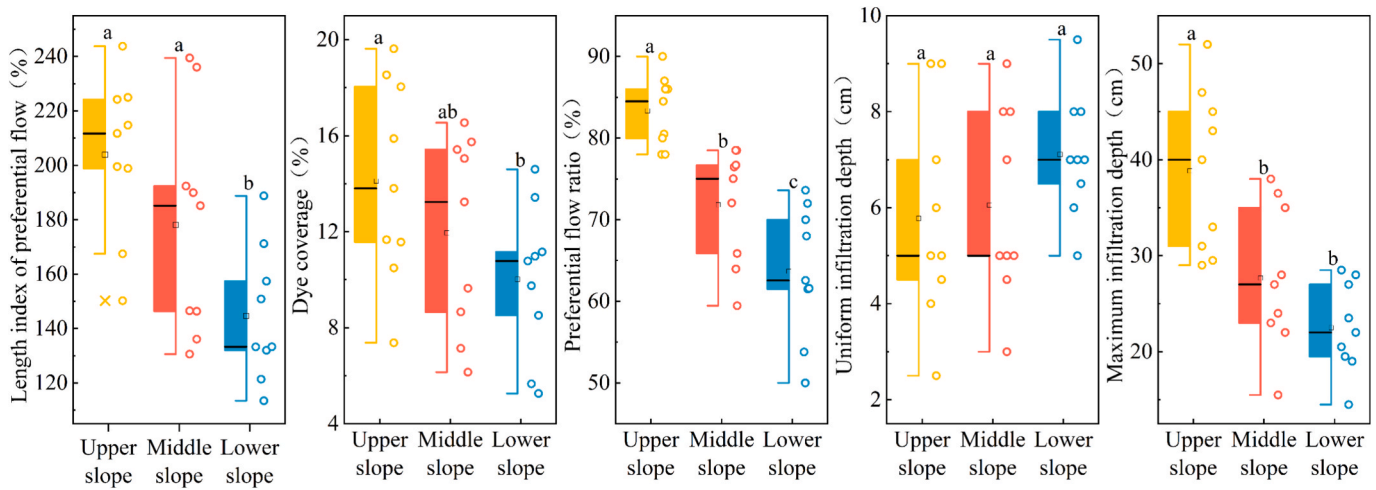


Fig. 5. Boxplots (maximum value, minimum value, median quartile, first quartile, and third quartile; dots indicate outliers; squares indicate the average value) of different preferential flow parameters in three slope positions ([a] length index of preferential flow, [b] dye coverage, [c] preferential flow ratio, [d] uniform infiltration depth, [e] maximum infiltration depth). Different letters mean significant difference at 0.05 level.

slope, with significant differences between the upper and lower slopes. Specifically, the upper slope exhibited a length index of $203.93 \pm 29.32\%$, surpassing $178.06 \pm 41.16\%$ of the middle slope and both significantly higher than $144.63 \pm 24.46\%$ of the lower slope. The upper slope had a D_c value of $14.11 \pm 4.18\%$, which was higher than the $11.95 \pm 4.06\%$ of the middle slope and significantly higher than the $10.01 \pm 3.15\%$ of the lower slope. Similarly, the P_f on the upper slope was $83.33 \pm 4.32\%$; which was significantly greater than the $71.83 \pm 7.03\%$ of the middle slope and the $63.67 \pm 8.08\%$ of the lower slope, representing increases of 1.16 and 1.32 times, respectively. Furthermore, the U_{max} was significantly greater on the upper slope at 38.83 ± 8 cm compared to the middle slope at 27.67 ± 7.11 cm and the lower slope at 22.5 ± 4.43 cm. However, this trend was notably inverted for the uniform penetration depth. The U_f was shallowest on the upper slope at 5.78 ± 2.08 cm, in contrast to the middle slope at 6.06 ± 1.89 cm and the lower slope at 7.33 ± 2.12 cm.

3.5. Influencing factors of preferential flow

3.5.1. Correlation analysis and dominant factors that facilitated preferential flow

A variety of physical properties exhibited correlations across different levels (Fig. 6). Specifically, the Li , P_f , and U_{max} variables demonstrated significant positive correlations with MCn , coarse rock fragments content (CCn), total rock fragments content (TCn), MCv , TCv , macroporosity (MPS), clay content, and bulk density ($P < 0.05$). In contrast, U_f showed very significant negative correlations with these influencing factors. Furthermore, Li and U_{max} correlated significantly negatively with silt content. Although root system indices and PF displayed no statistically significant relationship, a tentative negative trend was observed.

Similarly, Li , P_f , and U_{max} were all significantly positively correlated with the MCn and TCn (Fig. 7). While D_c was significantly positively correlated with the MCn . By contrast, U_f exhibited a highly significant negative correlation with the MCn and TCn and a significant negative correlation with CCn . Interestingly, no correlation was observed between $CCOn$ and PF characteristics parameters.

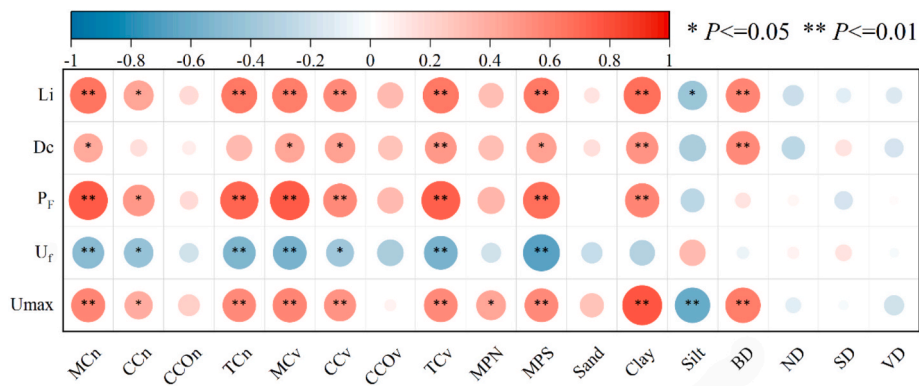


Fig. 6. Correlation coefficients among (a) rock fragments, conductivity, macropore, and soil physical properties and (b) among preferential flow as well as rock fragments, conductivity, macropore, and soil physical properties. The red colour represents positive correlations, and the blue colour represents negative correlations. The colour intensity of the blocks is proportional to the correlation coefficients. The legend colour shows the correlation coefficients and the corresponding colours. MCn : middle rock fragments content; CCn : coarse rock fragments content; $CCOn$: cobble rock fragments content; TCn : total rock fragments content; MCv : middle rock fragments coverage; CCv : coarse rock fragments coverage; $CCOv$: cobble rock fragments coverage; TCv : total rock fragments coverage; MPN : number of macropores; MPS : macroporosity; ND : number density of root; SD : surface area density of root; VD : volume density; Li : length index of preferential flow; D_c : dye coverage; P_f : preferential flow ratio; U_f : uniform infiltration depth; U_{max} : maximum infiltration depth.

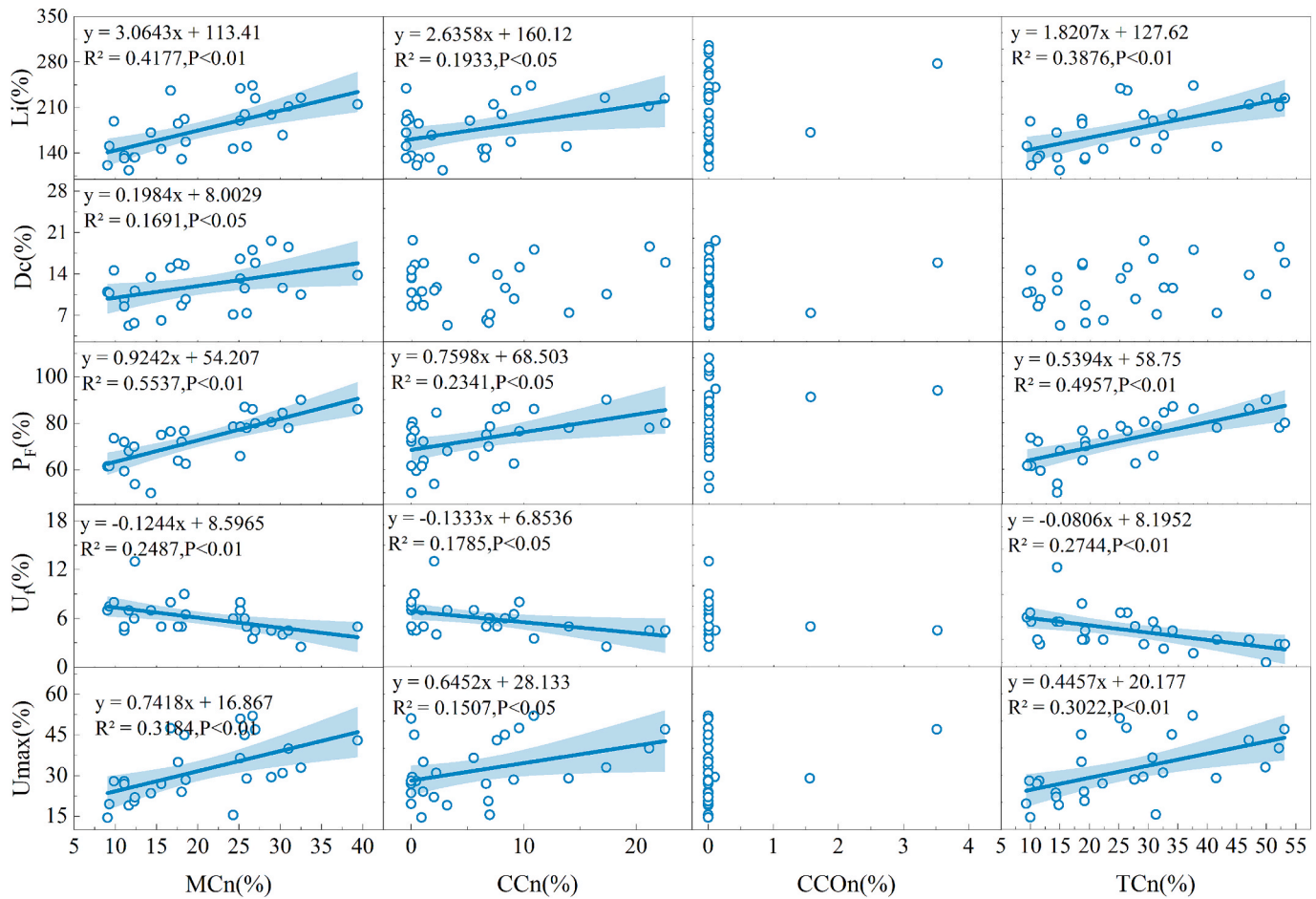


Fig. 7. Correlation between rock fragments content and preferential flow parameters. Li: length index of preferential flow; Dc: dye coverage; P_F: preferential flow ratio; U_f: uniform infiltration depth; U_{max}: maximum infiltration depth; MCn: middle rock fragments content; CCn: coarse rock fragments content; CCO: cobble rock fragments content; TCn: total rock fragments content.

To further quantify these relationships, a non-linear regression analysis was performed between rock fragment coverage and PF characteristics (Fig. 8). The analysis revealed that MCv was positively and significantly associated with Li, P_F, and U_{max}, and inversely with the U_f. Intriguingly, a non-linear, quadratic relationship was observed between CCv and Li, Dc, P_F, and U_{max}, with critical thresholds identified at 6.63 %, 6.42 %, 6.79 %, and 7.83 %, respectively, with a mean of 6.92 %. In a similar vein, TCv was found to be positively linked with Li, Dc, and P_F, yet inversely related to U_f.

3.5.2. PLSPM analysis

To clarify the intricate relationships between rock fragments, root systems, macropores, and PF, we constructed a partial least squares path model (PLSPM), as shown in Fig. 9. The model quantified path coefficients and their corresponding influence strengths, revealing that macropores exerted a highly significant direct effect on PF (path coefficient = 0.563). Notably, rock fragments had a substantially stronger impact on macropore formation than root systems, with path coefficients of 0.677 and -0.266, respectively. Furthermore, the path analysis suggested that the presence of rock fragments significantly promoted macropore formation, which in turn enhances PF. Therefore, it could be deduced that rock fragments were more pivotal in influencing soil structure and the facilitation of water movement.

4. Discussions

4.1. Why rock plays a more important role on PF at the rocky mountain area

For a long time, the response of PF to rocks and root systems and its impact on the groundwater hydrological processes in the North China mountainous areas with mixed soil and stone has been neglected. In these areas, rock fragments and root systems influence PF through distinct mechanisms, collectively shaping regional hydrological behavior. Our study demonstrates that rock fragments exert a stronger control on PF than root systems in these environments.

Rock fragments, integral to soil structure, promoted PF by creating a loose soil matrix with elevated non-capillary porosity and increased macroporosity (Dai et al., 2017; Zhang et al., 2023). This environment accelerated the velocity of water transport within soil profiles and enhanced lateral flow by fostering the connectivity of water flow channels (Lv et al., 2017; Zhu et al., 2020). Consequently, the development of interconnected macropores further facilitates PF (Liu et al., 2021; Su et al., 2023). As the content of rock fragments increased, so did the depth of preferential flow infiltration (Fig. 4). The high content of rock fragments prevented the surrounding soil from filling gaps, leading to the formation of large pores that facilitated PF (Nimmo, 2021; Sauer & Logsdon, 2002). At the upper slope, the preferential flow ratio, as well as the maximum infiltration depth, were significantly higher compared to those at the middle and lower slopes (Fig. 5). The rock fragments at these sites provided conduits for preferential flow along the soil–rock

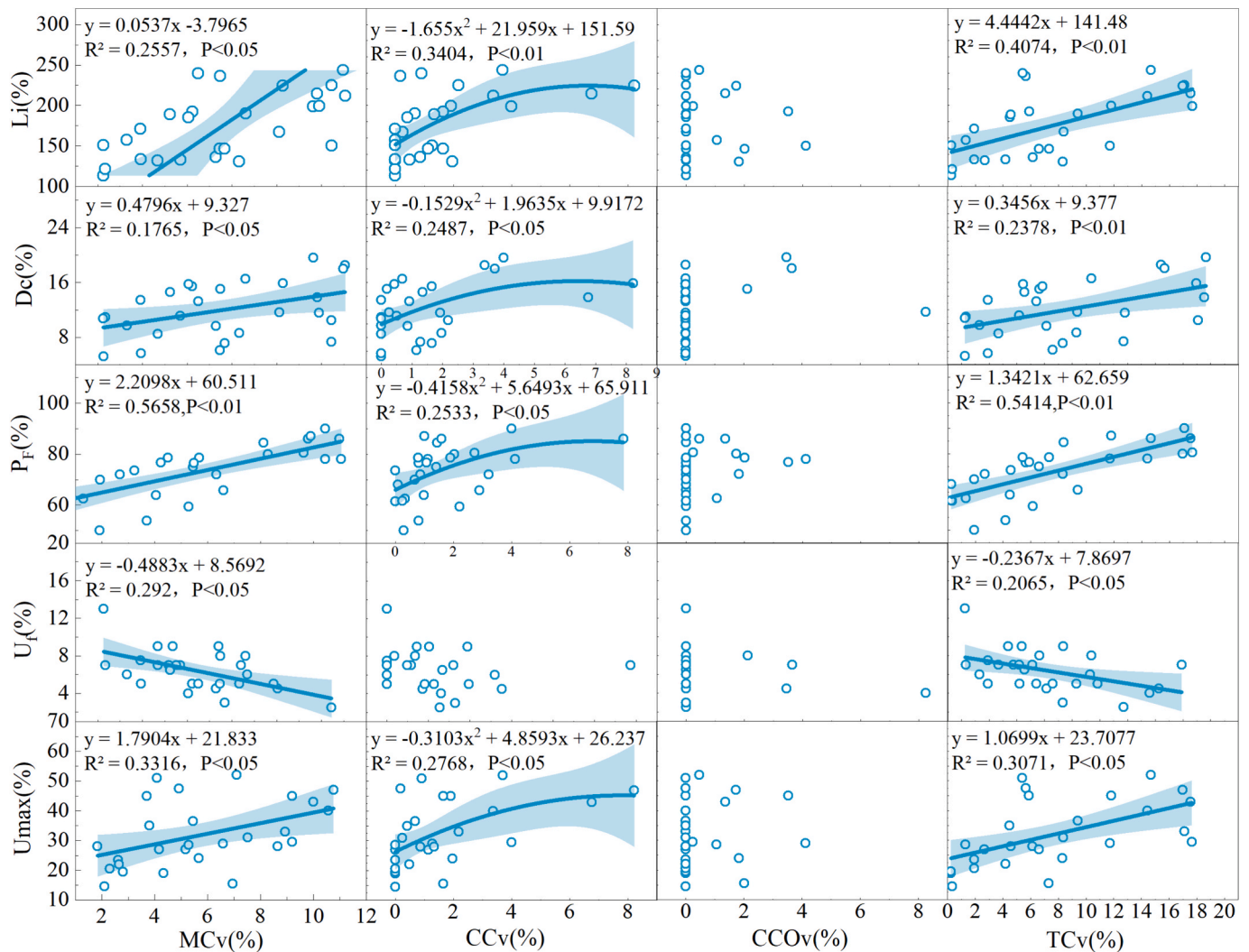


Fig. 8. Correlation analysis of rock fragments coverage and preferential flow characteristic parameters. Li: length index of preferential flow; Dc: dye coverage; Pf: preferential flow ratio; Uf: uniform infiltration depth; Umax: maximum infiltration depth; MCv: middle rock fragments coverage; CCv: coarse rock fragments coverage; CCOv: cobble rock fragments coverage; TCv: total rock fragments coverage.

interface.

The size of rock fragments significantly influenced PF formation, with medium rock fragments particles notably promoting PF more than coarse rock fragments, while cobbles did not show significant effect (Figs. 6, 7 and 8). This phenomenon primarily resulted from medium fragments' superior ability to promote macropore formation and improve water infiltration pathways (Dai et al., 2017). In contrast, larger rock fragments interacted less extensively with the soil matrix, reducing both macropore development and effective water flow cross-sections, ultimately creating complex and discontinuous infiltration patterns (Zhou et al., 2009).

The size of rock fragments played a significant role in PF formation, with medium-sized rock fragments (5–20 mm) notably promoting PF more than both smaller (2–5 mm) and larger (>75 mm) fragments (Figs. 6, 7 and 8). This can be explained by the dual effects of rock fragments on soil hydrological processes (Zhang et al., 2016). Medium-sized fragments create optimal interfacial spaces with the surrounding soil matrix, enhancing macropore connectivity while maintaining sufficient flow pathways (Zhou et al., 2011). Smaller fragments (<5 mm) tend to fill soil pores and reduce effective porosity (Wang & Shao, 2006), while larger fragments (>75 mm) decrease the available cross-sectional area for water flow and create more tortuous, discontinuous infiltration routes (Zhou et al., 2009). The weathering degree of medium fragments

also contributes to increased surface roughness and micro-fissures that facilitate preferential flow paths (Hou et al., 2023). These findings align with previous studies showing that medium rock fragments have the strongest positive correlation with macropore density (Luo et al., 2016) and infiltration rates (Wang et al., 2012).

Rock fragments coverage directly influenced PF formation by altering surface roughness and water infiltration pathways (Guo et al., 2010; Peng et al., 2011). When CCv was below 6.92 %, rock fragments covering the soil surface impeded flow development and inhibited uniform water infiltration (Fig. 8). The presence of rock fragments in the soil led to an increased curvature of water channels, compelled water to select specific pathways during infiltration, thereby promoting PF (Li et al., 2021). However, when CCv exceeded 6.92 %, it inhibited PF (Fig. 8). This is likely because high rock fragment content increased the tortuosity of water flow paths and reduced the effective cross-sectional area for infiltration (Zhang et al., 2016; Zhou et al., 2011). Additionally, the extensive surface coverage of rock fragments created physical barriers that disrupted continuous preferential flow pathways (Poesen et al., 1990; Tetegun et al., 2015).

Contrary to numerous studies suggesting that roots are significant drivers of soil PF (Ruan et al., 2022), our research indicated that in the rock fragments-rich regions of the rocky mountains in North China, roots did not predominantly drive soil PF. The limited spatial

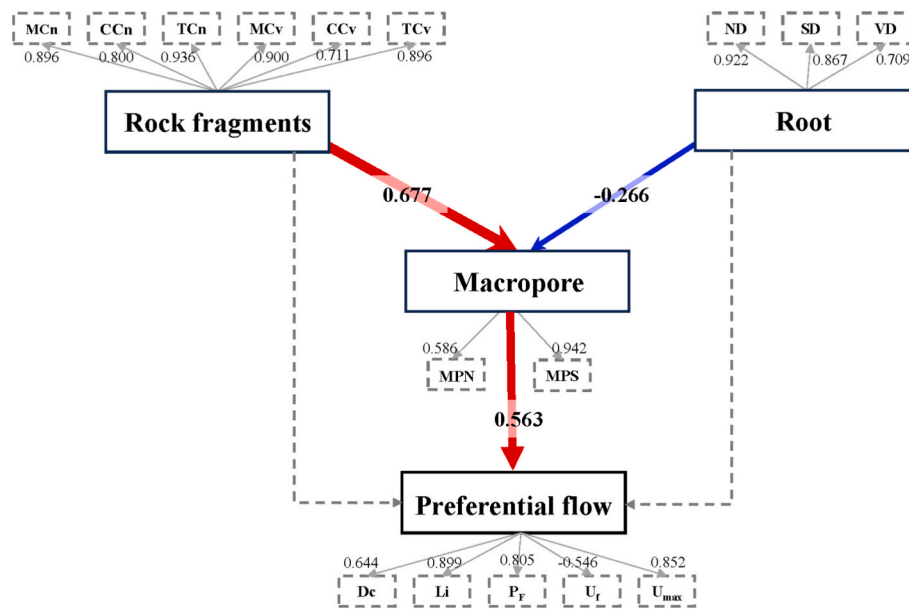


Fig. 9. Directed graph of the partial least squares path model (PLSPM). Each box represents an observed variable (i.e., measured). The dashed rectangle shows the load of the factors that produce the measured variable. Numbers show the standardized path coefficients (r_p). Path coefficients are calculated after 1000 bootstraps and reflected in the width of the arrow, with red and blue indicating positive and negative effects, respectively. Dashed arrows show that coefficients did not differ significantly from 0 ($P > 0.05$). The model is assessed using the Goodness of Fit (GoF) statistic, and the GoF value was 0.66. Abbreviation as previously stated. MCn: middle rock fragments content; CCn: coarse rock fragments content; CCO: cobble rock fragments content; TCn: total rock fragments content; MCv: middle rock fragments coverage; CCv: coarse rock fragments coverage; CCOv: cobble rock fragments coverage; TCv: total rock fragments coverage; MPN: number of macropores; MPS: macroporosity; ND: number density of root; SD: surface area density of root; VD: volume density; Li: length index of preferential flow; Dc: dye coverage; Pf: preferential flow ratio; Ur: uniform infiltration depth; Umax: maximum infiltration depth.

availability for root expansion in these areas compelled roots to grow adjacent to rock fragments to access nutrients and moisture (Palacio et al., 2014; Zwieniecki & Newton, 1994), resulting in clustered root distributions rather than extensive soil-wide networks (Estrada-Medina et al., 2013; Schwinning, 2013). This spatial constraint likely masked root-mediated flow dynamics. Moreover, the compressive effects of roots could diminish macropore sizes, resulting in a negative correlation with PF (Lucas et al., 2022). Preferential flow typically followed paths of least resistance, and in our study area, the high prevalence of rock fragments provided more favorable conditions for PF infiltration through the pores between rock fragments and soil, as well as through rock fissures, overshadowing the role of roots (Wang & Zhang, 2011). In the lower slope region, where most trees are artificially planted for 10–15 years, roots primarily serve to anchor and penetrate the soil, compressing larger pore spaces and hindering the development of decayed root channels that could act as PF pathways (Guerrero-Campo et al., 2006).

In conclusion, rock fragments played a more significant role in influencing preferential flow in the rocky mountain areas of North China due to their ability to create a network of macropores that facilitate rapid water movement. While root systems contribute to soil structure and water uptake, their impact on preferential flow is less pronounced in comparison. Understanding the interplay between rock fragments and root systems is essential for managing water resources and maintaining ecosystem resilience in these regions, particularly as climate conditions continue to evolve.

4.2. The importance of rock fragments-induced preferential flow to the ecosystems of the rocky mountain in North China

Our study highlights the critical role of rock fragment-induced PF in shaping North China Rocky Mountain ecosystems. Rock fragments, as a significant component of soil structure, contributes notably to the formation of macropores, which are essential for facilitating water movement (Hou et al., 2023; Zhang et al., 2016). This mechanism proves

particularly vital for maintaining hydrological balance in regions with erratic precipitation and water scarcity.

The presence of rock fragments improved water infiltration (Jomaa et al., 2012), and reduced surface runoff (Jarvis, 2020; Wang et al., 2012), thereby enhancing the availability of moisture and nutrients to plant roots and supporting plant growth. However, the optimal rock fragments content in soil should be balanced to ensure that the benefits of PF are realized without compromising other soil functions.

Furthermore, the role of rock fragments in macropore formation is more pronounced than that of root systems, suggesting that managing soil rock fragments content could positively influence ecosystem resilience when appropriately calibrated. Strategic rock fragments content management might support biodiversity by ensuring adequate water availability for native flora and fauna, thus fostering a more sustainable ecosystem. In conclusion, rock fragments-induced PF is a pivotal process in the water dynamics of the North China Rocky Mountains. As climate change progresses, understanding and regulating this process will be paramount for ecosystem sustainability, requiring careful optimization rather than arbitrary increases in rock fragment content.

Overall, our results highlight the pivotal role of rock-fragment-induced preferential flow within the rocky mountain landscape of North China. The observed thresholds (6.92 % for coarse-gravel coverage) provide quantitative guidance for land managers seeking to optimise water infiltration while minimising runoff. However, the present inferences are derived from a single, low-intensity irrigation event (5.5 mm at 1.25 mm h⁻¹) conducted on two slopes under moderate antecedent moisture. Such gentle wetting may over-enhance surface retention in low-rock plots (Bao et al., 2024; Yin et al., 2025) and could underestimate deep preferential flow during the high-intensity convective storms typical of the region. These simplifications mean that the reported dominance of rock fragments over roots in driving PF applies strictly to the tested irrigation regime. Future campaigns should therefore apply a gradient of rainfall energies to bracket the full hydrological response space. Only through such multi-condition observations can the quantitative thresholds presented here be validated and extrapolated to

entire catchments under a changing climate.

5. Conclusion

In the rocky mountainous regions of North China, our study employed dye tracing experiments to quantitatively analyze preferential flow (PF). We observed that rock fragments content and coverage increased with slope elevation, with small rock fragments particles predominantly composing the soil at each position. In contrast, root characteristics, including quantity density, surface area density, and volume density, were more abundant in the lower slope. Soil macropore density decreased as the slope descended, with better connectivity observed in the upper slope. The preferential flow characteristics exhibited a decreasing trend with increasing soil depth, with the upper slope demonstrating significantly higher values compared to the middle and lower slopes. Rock fragments particles between 5 and 20 mm showed the most significant positive correlation with PF. Rock fragments coverage thresholds of 6.92 % for coarse rock fragments indicated that PF was inhibited beyond these levels. The presence of rock fragments may diminish the beneficial effects of plant roots on PF. This insight highlights the pivotal role of rock fragments in influencing PF in rocky mountain areas, with significant implications for water resources management in arid regions.

CRedit authorship contribution statement

Jianfei Li: Writing – original draft, Software, Data curation. **Jinhao Zhang:** Software, Data curation. **Yuefeng Shi:** Supervision. **Shan Lu:** Investigation. **Wu Tang:** Investigation. **Jinsong Zhang:** Methodology. **Si-Yi Zhang:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Zhi-Hua Zhang:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.catena.2025.109560>.

Data availability

[preferential flow \(Original data\)](#) (Mendeley Data)

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